



NATIONAL CENTRE OF
ROBOTICS & AUTOMATION

INTERNSHIP REPORT

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Submitted To:

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Acknowledgment

I am sincerely grateful to the **National Centre of Robotics and Automation (NCRA)** for providing me with the opportunity to intern at such a forward-thinking organization. This internship allowed me to gain hands-on experience in both PCB designing and AI, which has been instrumental in enhancing my technical skills.

During the internship, I worked on designing a modified Arduino board in the PCB Designing Department and explored the field of computer vision in the AI Department, where I focused on the OpenCV library and model training. These experiences have significantly contributed to my professional growth.

I would also like to extend my heartfelt thanks to **Mr. Ibtehaj, Mr. Abdul Rahim, Mr. Hassan Akbar**, and the entire team at NCRA for their guidance and support. As a student of the **Pakistan Institute of Engineering and Applied Sciences (PIEAS)**, I am proud to have had the opportunity to apply my academic knowledge in such a practical and impactful way.

Abstract

This report details the comprehensive experience gained during my internship at the National Centre of Robotics and Automation (NCRA), focusing on two key areas: PCB designing and computer vision using the OpenCV library. Over the course of six weeks, I undertook a diverse set of tasks that allowed me to bridge theoretical knowledge with practical application.

In the span of my six-week internship at NCRA, I mainly worked with PCB designing and Computer vision. During the first four weeks, I worked on a modified version of an Arduino UNO board; working from Schematic design to Component library generation and finally PCB layout. This work helped me to improve my electronics and design skills. I wrapped up the last couple of weeks learning about computer vision with OpenCV, which further solidified my knowledge of AI that day by covering image manipulation and object tracking. Overall, the internship was a formative experience and cemented my roots in hardware design as well as artificial intelligence.

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1. Introduction

The **National Centre of Robotics and Automation (NCRA)** is a leading research and development institution in Pakistan, established to spearhead technological innovation in robotics and automation. With a vision of enabling highly skilled experts to foster an ecosystem focused on indigenous technology development, NCRA aims to make Pakistan one of the world's technologically advanced nations. The center's mission is to support researchers, scientists, and experts by creating a consortium of advanced labs across the country, ensuring that research and development activities align with NCRA's overarching vision.

NCRA is driven by several key objectives, including building national capacity in robotics and automation through tackling 'hard' problems identified by a scientific committee of local and international experts. The center is also committed to solving local challenges and commercializing technology solutions, providing high-value shared services to academia, startups, and industry partners, and contributing to human resource development in robotics and automation. These goals reflect NCRA's dedication to not only advancing technological research but also ensuring that these advancements have a practical impact on society.



Figure 1: National Centre of Robotics and Automation

2. Distribution of NCRA

The National Centre of Robotics and Automation is a consortia of 11 labs over 13 universities in Pakistan with its center headquarters at NUST College of E&ME.

Robot Maker Lab: The Robot Maker Lab is equipped with state-of-the-art rapid prototyping, machining, and fabrication facilities for robot parts. The Lab has two main streams dealing with mechanical and electrical part manufacturing of robots.

Robot Design and Development: The primary objective of the lab is to provide a fusion center where real-world (preferably local) problems are picked, given to a multidisciplinary team, and led to commercialization.

Human-Centered Robotics: The Human-Centered Robotics Lab aims to indigenously design and develop integrated robotic systems, based on smart sensing and actuation, to seamlessly interact with humans, actively learn from them and eventually create an effective collaborative environment.

Industrial Monitoring and Automation: The lab aims for industrial automation and control systems. It has scope in induction motor power, health monitoring systems, motor characterization and improvement facility.

MEM's Sensor Design and Testing: The lab develops skilled manpower through scientific research to produce world-class MEM's experts having expertise in design and testing to produce world-class R&D in MEM's and initiate commercial activities in this area in Pakistan.

Agricultural Robotics: The lab aims to realize the potential of precision agriculture and agricultural robotics technologies for higher water efficiency, enhanced agricultural productivity and environmental sustainability via peer-reviewed research, industrial collaborations, prototyping new systems, creation of testbeds, job creation, trainings and entrepreneurship.

UAV Dependability: This lab aims to address the challenges in Unmanned Aircraft Systems (UAS) development. It is a major step towards self-reliance and indigenous development of UAS.

Swarm Robotics: This lab aims to establish a scaling solution for National Centre of Robotics and Automation by providing an application-agnostic swarm intelligence platform. The lab is composed of three sub-labs, i.e., Swarm Intelligence Lab, Computer Vision and AI Lab, and Decentralized Communications Lab.

Haptics, Human-Robotics & Condition Monitoring: This lab is the joint lab of NEDUET and MUET, whose aim is to execute the state-of-the-art researches and development in the field of Haptics & Human- Robotics Lab (HHR) at NED University and Condition Monitoring Systems (CMS) at Mehran University.

Control Automotive and Robotics: The Scope of the Lab is to develop Intelligent Social, and Search and Rescue Robotics, and autonomous vehicles; utilizing Machine Learning and Deep Learning. This lab aims to facilitate high quality original scientific research in robotics with direct participation of artificial agents in search and rescue applications.

Advanced Robotics and Automation: As part of a response to our national challenges such as lack of agriculture technology and productive automation, the lab is involved in research and prototype development pertaining to all aspects of precision agriculture, industrial automation and biomedical devices.

3. Robot Maker Lab

The **Robot Maker Lab**, operating under the National Centre of Robotics and Automation (NCRA) at the NUST College of EME, is a cutting-edge manufacturing facility offering specialized design and production services in both Electrical and Mechanical Engineering domains. This lab serves as a crucial resource, providing state-of-the-art solutions to academia, startups, industry, and other NCRA labs by acting as a centralized hub for robot manufacturing and prototyping.

Aim and Scope

The Robot Maker Lab is equipped with advanced rapid prototyping, machining, and fabrication tools specifically designed for the creation of robot parts. It features two primary streams, one focused on mechanical component production and the other on electrical systems, including PCB circuit design and fabrication. This facility provides comprehensive support to NCRA labs and external partners, offering services such as metalworking, plastic and wood processing, carbon fiber fabrication, and more.

Major Facilities

The lab houses a range of high-end technologies including:

- A 5-axis CNC machining center for precision manufacturing,
- A state-of-the-art 3D printing center,
- Laser-based precision machining,
- 3D scanning for reverse engineering,
- Grinding, polishing, and welding stations,
- PCB prototyping and testing capabilities.

Target Outcomes and Beneficiaries

With a target revenue of PKR 9.5 million over the next three years, the Robot Maker Lab aims to foster job creation and support the commercialization of research. The lab aims to conduct workshops to train over 100 individuals and offer consultancy services to local industries and startups. Beneficiaries of the lab include NCRA labs, researchers in robotics, and various industry partners looking to leverage cutting-edge technologies for rapid product development and prototyping.

4. Objectives

The primary objectives of my internship at the National Centre of Robotics and Automation (NCRA), specifically at the Robot Maker Lab, were designed to align both with my academic goals as an Electrical Engineering student and the overarching goals of the organization. These objectives focused on the enhancement

of my technical skills, practical experience, and professional development in the fields of PCB design and computer vision, while also contributing to the lab's mission of fostering technological advancements in Pakistan. The following were the key objectives I set out to achieve:

4.1 Academic Objectives

- **Enhancement of Technical Skills in PCB Design:** A major objective was to deepen my understanding and practical knowledge of PCB (Printed Circuit Board) design, including schematic creation, library design, component placement, layout, and the preparation of manufacturing documentation. This objective was specifically aligned with my academic studies in electrical engineering, as PCB design is a critical skill in circuit design and embedded systems.
- **Development of Computer Vision Expertise:** I aimed to gain exposure to the field of computer vision by working with the OpenCV library and applying machine learning techniques. This would help me bridge the gap between theoretical coursework and practical applications of image processing, object detection, and pose estimation. The goal was to apply these concepts to real-world problems and broaden my understanding of artificial intelligence (AI) in automation systems.
- **Application of Engineering Principles in Real-World Projects:** I sought to integrate the engineering principles I had learned in my academic coursework into practical, hands-on projects. This objective was to solidify my theoretical understanding of circuit design and machine learning by applying them to real-world tasks in the lab.

4.2 Professional Objectives

- **Gain Practical Experience in a Real-World Engineering Environment:** My goal was to gain valuable experience working in a professional engineering environment, which would help me understand the complexities of project management, teamwork, and collaboration. I wanted to familiarize myself with the workflow and culture of a research and development lab, learning how to manage engineering projects from concept to completion.

- **Understand Industry-Level PCB Design and Manufacturing Processes:** A key objective was to learn the industrial standards and procedures involved in PCB design and manufacturing, including the use of Altium Designer, and how to create documentation required for PCB fabrication. This objective was designed to give me insight into the practical challenges of transitioning from design to production.
- **Contribute to Research and Development at NCRA:** I aimed to contribute meaningfully to the lab's ongoing research and development efforts, particularly in the areas of robotics and automation. By working on PCB design projects and computer vision tasks, I sought to support the lab's mission of developing indigenous technologies and solving local problems through innovation.

4.3 Personal Development Objectives

- **Improve Problem-Solving and Critical Thinking Skills:** I set out to enhance my problem-solving abilities by working on complex engineering challenges, both in PCB design and computer vision. I wanted to improve my critical thinking skills by identifying issues and proposing effective solutions, whether they were related to design, software development, or project implementation.
- **Enhance Communication and Teamwork Skills:** Another objective was to improve my ability to work as part of a multidisciplinary team. This included developing clear communication skills, both written and verbal, as well as learning how to collaborate with engineers, researchers, and supervisors on project tasks.
- **Time Management and Project Coordination:** I aimed to improve my time management and project coordination skills by balancing multiple tasks and meeting deadlines. This objective was crucial in learning how to manage engineering projects within time constraints and under supervision.

By achieving these objectives, my internship was designed to not only deepen my technical and professional skills but also prepare me for future challenges in my academic and professional career in electrical engineering, with a particular focus on automation, robotics, and machine learning.

5. Internship Details

5.1 Work Environment

During my internship at the **National Centre of Robotics and Automation (NCRA)**, specifically at the **Robot Maker Lab**, I was exposed to an advanced technological environment that provided hands-on experience in PCB design and Computer Vision. The lab's infrastructure included cutting-edge tools such as CNC machines, laser cutters, 3D printers, and PCB manufacturing equipment. The workspace was designed to foster collaboration, innovation, and practical learning.

I was introduced to the lab's facilities during an orientation session, which helped me understand the workflows and projects being executed. The lab was divided into various sections based on the projects, such as PCB fabrication, computer vision research, and 3D prototyping. I was assigned a desk in the PCB design section, where I worked closely with engineers and researchers.

My supervisor, along with the rest of the team, guided me through the initial phase of understanding the lab's objectives, including the application of technology to solve real-world problems through automation and robotics. The collaborative nature of the work environment helped me adapt quickly and contribute meaningfully to the projects.

5.2. Internship Objectives

The primary objectives set for my internship were to:

- Gain hands-on experience in PCB design and fabrication, including schematic design, component placement, and layout creation.
- Develop and apply computer vision techniques to real-world problems using **OpenCV** and other relevant libraries.
- Understand the full pipeline of PCB manufacturing, from schematic design to prototyping and final production.

- Learn the fundamentals of machine learning and its application in the field of computer vision, including object detection and pose estimation.
- Work with teams to develop prototypes that align with NCRA's vision of creating indigenous robotic technologies.

The work environment encouraged open communication and teamwork, which allowed me to exchange ideas with engineers, developers, and other interns. The collaborative atmosphere enabled me to complete my assigned tasks effectively while broadening my knowledge in robotics and automation technologies.

6. Description of Work Done

6.1. PCB Designing

During the first four weeks of my internship, I worked extensively on PCB designing. The process involved the following detailed steps:

6.1.1 Step 1: Schematic Design

The schematic design is the blueprint of the PCB, where all the components are connected logically. This involves selecting the appropriate components and defining how they will interact. The software used for the design was Altium Designer.

1. **Component Designing:** I began by designing all the components by myself, including resistors, capacitors, ICs, and connectors.
 - Key components used: Microcontroller (e.g., ATmega328), Voltage Regulators, Type B Connector for USB.
 - For each component, I ensured that accurate and perfect footprints were created.

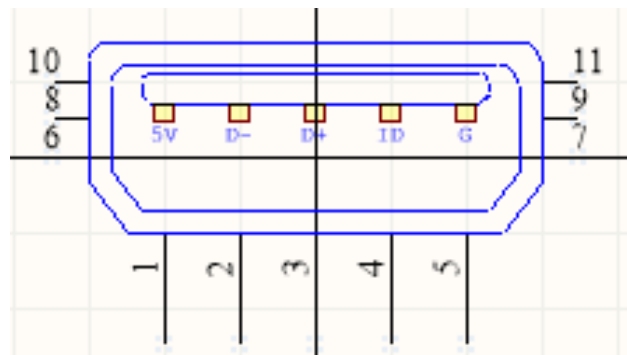


Figure 2: USB Connector Schematic

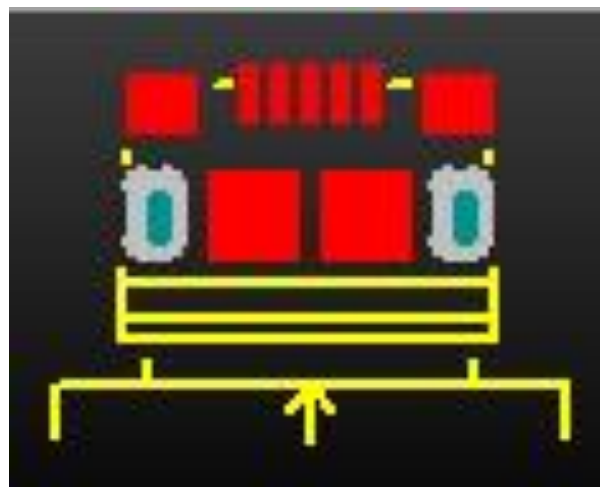


Figure 3: USB Connector Footprint

2. **Pin Configuration:** Once the components were placed, I connected them according to the circuit logic. This step was crucial in ensuring that all components interacted correctly.
 - Example: Connecting the power pins of the microcontroller to the appropriate voltage source and grounding pins.

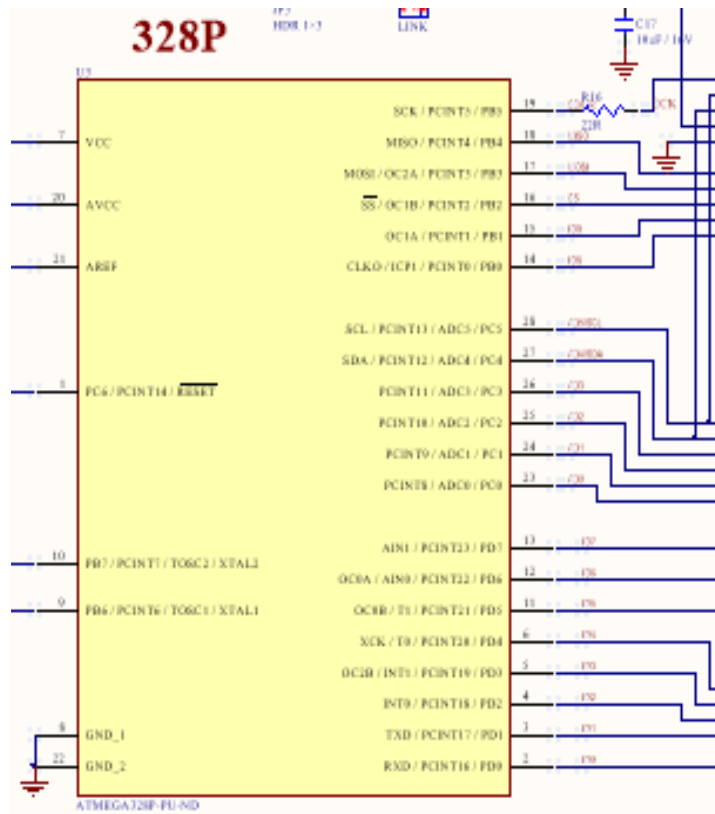


Figure 4: 328P Schematic

6.1.2 Step 2: PCB Placement

Once the schematic was complete, the next step was to place the components on the PCB layout.

1. **Placing Components:** Each component was placed in its designated area. I had to ensure that the power components (e.g., voltage regulator) were close to the power input and the microcontroller was centrally located.



Figure 5: PCB Component Placement

2. **Power and Ground Planes:** A critical aspect of the design was creating a solid ground plane and defining the power plane. This ensures minimal noise and interference in the PCB.

6.1.3 Step 3: PCB Layout

After the component placement, the PCB layout was designed, where the routing of connections between components was performed.

1. **Trace Routing:** I routed the traces between the pins of the components, making sure to follow best practices like keeping signal traces short and avoiding crossing traces unnecessarily. For high-power lines, I used thicker traces.
 - Example: The trace width for power lines was set to 0.4mm, while signal traces were kept at 0.3mm.

[Placeholder for PCB layout image]

2. **Via Placement:** Whenever a signal needed to be routed between layers, vias were placed to connect the traces across different layers.

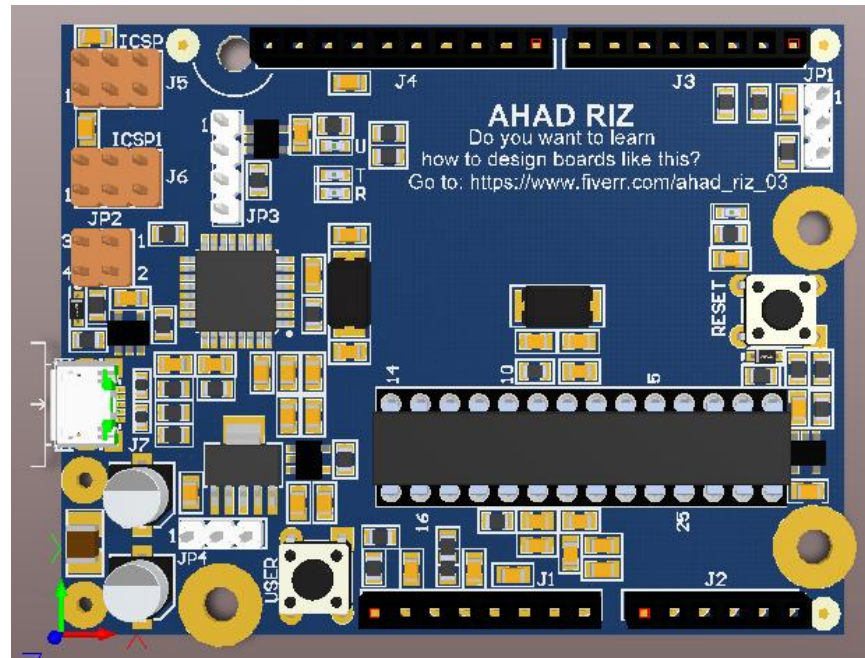


Figure 6: Final PCB Layout

6.1.4 Step 4: Manufacturing Documentation

Once the layout was complete, I generated the necessary files for manufacturing.

1. **Gerber Files:** The Gerber files were created, which are the industry standard for PCB manufacturing. These files include the layers for copper, solder mask, and silkscreen.
2. **Bill of Materials (BOM):** The BOM was generated, listing all the components used, their values, and footprints. This is crucial for ordering the components.

Component list													
28Pins													
Source Data From: PCB_Project.PjgPak				Contact:									
Project: PCB_Project.PjgPak													
Variant: Base													
Report Date: 17/12/2024				11:41:24 AM									
Print Date: 12-Sep-24				18:24:24 AM									
#	Category	Manufacturer 1	Manufacturer 2	Package / Case	Description	Quant Qty	Supplier 1	Supplier Part Number 1	Suppl for Order # 421	Suppl for Stock # 421	Suppl for Mail # 421	Suppl for Subst # 421	Suppl for Comm # 421
1	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
2	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
3	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
4	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
5	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
6	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
7	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
8	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
9	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
10	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
11	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
12	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
13	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
14	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
15	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
16	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
17	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
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33	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
34	Capacitors	Leadless	Leadless	0805 (2012 Metric)	High Voltage Ceramic Capacitor, 10	2	Digikey	445-2545-1-ND					Leading
Subtotal				10 pcs: Total 199.67 <none>									
				Price per Item 19.97 <none>									

Figure 7: Bill of Materials

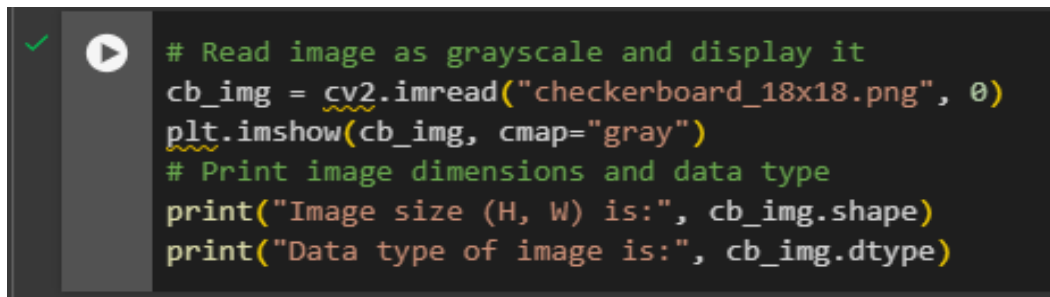
6.2 Computer Vision Using OpenCV

During the final two weeks of my internship, I worked extensively with the OpenCV library, which is a foundational tool for Computer Vision applications. Below is a detailed account of the tasks I performed and the concepts I covered during this period.

6.2.1 Getting Started with Images

In this section, I began by loading, displaying, and manipulating images using OpenCV and other libraries like NumPy and Matplotlib. The core functionality of OpenCV allowed me to read images as arrays of pixel values, which I could modify and visualize.

- **Image Loading and Displaying:** I loaded images such as checkerboard patterns and logos, displaying them with `cv2.imread()` and `matplotlib`.
- **Data Inspection:** I printed image data, including pixel values, dimensions, and types, to understand how OpenCV stores image data.



```
# Read image as grayscale and display it
cb_img = cv2.imread("checkerboard_18x18.png", 0)
plt.imshow(cb_img, cmap="gray")
# Print image dimensions and data type
print("Image size (H, W) is:", cb_img.shape)
print("Data type of image is:", cb_img.dtype)
```

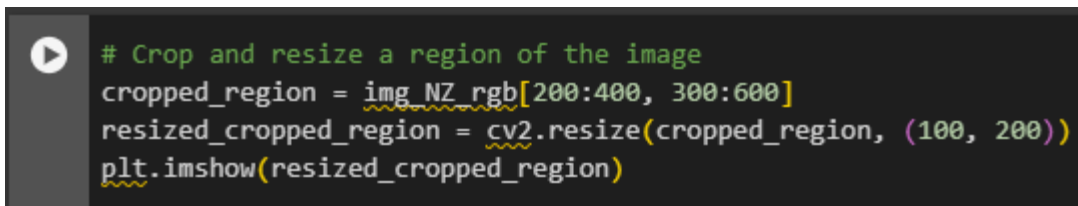
Figure 8: Starting with images code

This section laid the foundation for the advanced operations performed later on in the project.

6.2.2 Basic Image Manipulation

I worked on basic image manipulation, such as accessing pixel values, modifying images, and applying cropping and resizing. This gave me an understanding of how to manipulate image data directly and how to resize images while maintaining aspect ratios.

- **Cropping and Resizing:** Cropping involved selecting a specific region of an image and resizing it using OpenCV's `resize()` function.



```
# Crop and resize a region of the image
cropped_region = img_NZ_rgb[200:400, 300:600]
resized_cropped_region = cv2.resize(cropped_region, (100, 200))
plt.imshow(resized_cropped_region)
```

Figure 9: Code for image manipulation

This section taught me how to work with different parts of an image and how resizing affects image quality.

6.2.3 Annotating Images

Next, I learned how to annotate images by drawing shapes and adding text. This is useful in object detection tasks where we highlight regions of interest with rectangles, circles, or text annotations.

- **Drawing Lines, Circles, and Rectangles:** I used functions like `cv2.line()`, `cv2.circle()`, and `cv2.rectangle()` to add visual annotations on images.
- **Adding Text:** I explored text overlay on images using `cv2.putText()`.

```
# Draw a rectangle and add text
cv2.rectangle(image, (500, 100), (700, 600), (255, 0, 255), thickness=5)
cv2.putText(image, "Apollo 11", (200, 700), cv2.FONT_HERSHEY_PLAIN, 2.3, (0, 255, 0), 2)
plt.imshow(image[:, :, ::-1])
```

Figure 10: Code for annotating images

Annotating images is crucial in marking important areas, especially in object detection and feature extraction tasks.

6.2.4 Image Enhancement

In the image enhancement section, I experimented with adjusting the brightness and contrast of images. I used arithmetic operations like `cv2.add()` and `cv2.subtract()` for brightness changes and `cv2.multiply()` for contrast adjustments.

- **Brightness Adjustment:** I increased and decreased brightness by adding or subtracting constant values from the pixel values.
- **Contrast Adjustment:** I multiplied the pixel values by factors to modify the contrast.

```
# Increase brightness
matrix = np.ones(img_rgb.shape, dtype="uint8") * 50
img_rgb_brighter = cv2.add(img_rgb, matrix)
plt.subplot(133); plt.imshow(img_rgb_brighter); plt.title("Brighter");
```

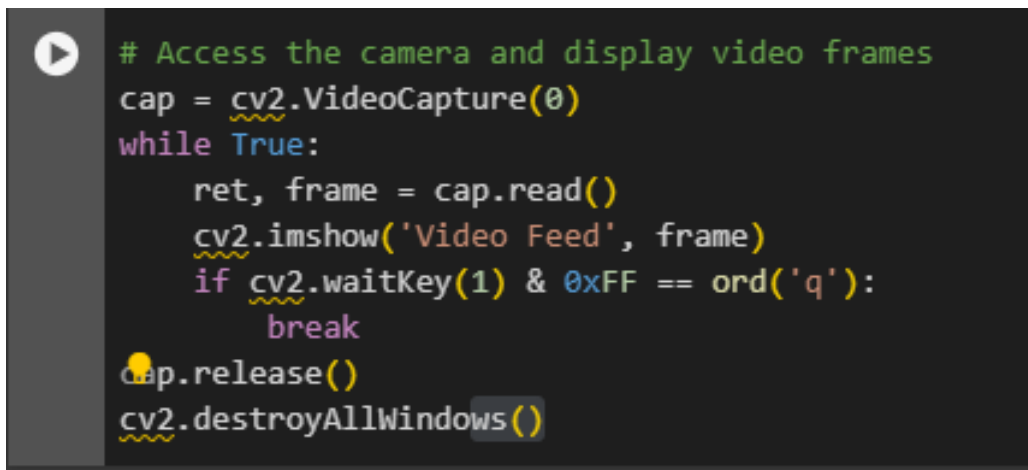
Figure 11: Code for image enhancement

These techniques allowed me to understand how pixel values control the overall appearance of images.

6.2.5 Accessing the Camera

The final task in this part was accessing the camera using OpenCV. I captured real-time video from the webcam and displayed it frame by frame, which laid the groundwork for object tracking and face detection tasks in later stages.

- **Capturing Video:** I used `cv2.VideoCapture()` to access the webcam and `imshow()` to display the video feed in real-time.



```
# Access the camera and display video frames
cap = cv2.VideoCapture(0)
while True:
    ret, frame = cap.read()
    cv2.imshow('Video Feed', frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

Figure 12: Code for accessing camera

This section marked the transition from working with static images to handling dynamic, real-time data.

6.2.6 Image Filtering and Manipulation

One of the fundamental tasks I worked on was image filtering and manipulation. I implemented various filters such as Canny Edge Detection, Blurring, and Corner Detection using OpenCV. A basic implementation to toggle between these filters is shown below.

```
# Filters applied on live camera feed
if image_filter == PREVIEW:
    result = frame
elif image_filter == CANNY:
    result = cv2.Canny(frame, 80, 150)
elif image_filter == BLUR:
    result = cv2.blur(frame, (13, 13))
elif image_filter == FEATURES:
    frame_gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
    corners = cv2.goodFeaturesToTrack(frame_gray, **feature_params)
    if corners is not None:
        for x, y in numpy.float32(corners).reshape(-1, 2):
            cv2.circle(result, (x, y), 10, (0, 255, 0), 1)
```

Figure 13: Code for image filtration and manipulation

6.2.7 Image Feature Matching and Alignment

I also worked on aligning scanned images to a reference using feature matching techniques. The **ORB (Oriented FAST and Rotated BRIEF)** detector was used to extract keypoints and match them between two images. A **homography transformation** was then applied to align the scanned image with the reference image.

```
# Detect ORB features and compute descriptors.
orb = cv2.ORB_create(MAX_NUM_FEATURES)
keypoints1, descriptors1 = orb.detectAndCompute(im1_gray, None)
keypoints2, descriptors2 = orb.detectAndCompute(im2_gray, None)

# Match features and sort
matches = matcher.match(descriptors1, descriptors2, None)
matches.sort(key=lambda x: x.distance, reverse=False)

# Apply homography
h, mask = cv2.findHomography(points2, points1, cv2.RANSAC)
im2_reg = cv2.warpPerspective(im2, h, (width, height))
```

Figure 14: Code for image matching and alignment

6.2.8 Panorama Stitching

I created a panorama using a set of overlapping images. By using OpenCV's stitching module, I was able to automatically align and stitch multiple images into a single wide-field panorama.

```
stitcher = cv2.Stitcher_create()
status, result = stitcher.stitch(images)

if status == 0:
    plt.imshow(result)
```

Figure 15: Code for panorama stitching

6.2.9 High Dynamic Range (HDR) Imaging

To enhance the visual quality of images, I implemented HDR Imaging using multiple images taken at different exposure levels. The **Debevec CRF** (Camera Response Function) was used to merge the images and create an HDR image, which was then tone mapped using different methods.

```
mergeDebevec = cv2.createMergeDebevec()
hdrDebevec = mergeDebevec.process(images, times, responseDebevec)

# Tonemapping using Drago's method
tonemapDrago = cv2.createTonemapDrago(1.0, 0.7)
ldrDrago = tonemapDrago.process(hdrDebevec)
```

Figure 16: Code for HDR imaging

6.2.10 Object Tracking

Using various tracking algorithms such as KCF and CSRT, I developed an object tracking system. This system was able to track a specified object throughout a video and maintain a bounding box around it.

```
▶ tracker = cv2.TrackerKCF_create()
  ok = tracker.init(frame, bbox)

  while True:
      ok, bbox = tracker.update(frame)
      if ok:
          drawRectangle(frame, bbox)
      else:
          drawText(frame, "Tracking failure detected", (80, 140), (0, 0, 255))
```

Figure 17: Object tracking code

6.2.11 Face Detection

Lastly, I implemented face detection using a **Caffe-based deep learning model**. The model was used to detect faces in real-time video streams.

```
▶ net = cv2.dnn.readNetFromCaffe("deploy.prototxt", "res10_300x300_ssd_iter_140000")
  # Processing each frame for face detection
  blob = cv2.dnn.blobFromImage(frame, 1.0, (300, 300), (104.0, 177.0, 123.0))
  net.setInput(blob)
  detections = net.forward()
```

Figure 18: Code for face detection

7. Challenges Faced

7.1 Technical Challenges

Both in PCB design and computer vision, I faced technical hurdles such as managing complex routing in the PCB layout and optimizing object detection models for real-time performance.

7.2 Time Management Challenges

Given the limited duration of the internship, balancing multiple tasks required efficient time management, especially during the transition from PCB design to computer vision.

8. Skills Gained

8.1 Technical Skills

- **PCB Designing:** Gained proficiency in Altium Designer, schematic creation, and layout optimization.
- **Computer Vision:** Developed skills in image processing, object detection, and pose estimation using OpenCV and TensorFlow.

8.2 Soft Skills

- **Problem Solving:** Developed critical problem-solving abilities while troubleshooting PCB issues and improving image processing models.
- **Communication:** Collaborated effectively with mentors and peers, improving both written and verbal communication.

9. Conclusion

My internship at the National Centre of Robotics and Automation (NCRA) was a transformative learning experience, offering hands-on exposure to both **PCB design** and **computer vision** technologies. The first half of the internship allowed me to work extensively on PCB design, where I designed and developed a modified version of the Arduino UNO, making it more adaptable for various voltage levels and upgrading its connectivity with a Type B connector. The second half focused on computer vision, where I worked with **OpenCV**, implementing various tasks such as image manipulation, object tracking, and pose estimation. This internship not only honed my technical skills but also deepened my understanding of how engineering principles apply to real-world scenarios, bridging the gap between academic learning and practical application. Moreover, the diverse nature of tasks allowed me to develop problem-solving skills and adaptability in different technological areas. These experiences have strengthened my confidence in pursuing further challenges in the fields of **robotics, automation, PCB design, and computer vision**.

Recommendations

Based on my internship experience at NCRA, I have a few recommendations for enhancing the internship program for future participants:

- **Advanced PCB Design Techniques:** While the foundational knowledge provided was strong, future interns could greatly benefit from more detailed training in advanced PCB design techniques. Specific areas such as multi-layer PCB design, advanced routing methods, and thermal analysis would allow interns to tackle more complex projects and gain an even deeper understanding of PCB manufacturing processes.
- **Comprehensive Computer Vision Training:** In the computer vision segment, the inclusion of a more structured introduction to advanced topics such as **deep learning integration with OpenCV** or **TensorFlow** could help interns explore the cutting-edge applications of computer vision. This would prepare them to work on more complex object detection, recognition, and tracking tasks that go beyond the basics of OpenCV.
- **Cross-Disciplinary Learning:** Providing opportunities for interns to collaborate across disciplines (e.g., PCB designers working with computer vision engineers) could lead to innovative solutions that combine hardware and software expertise. Interns can benefit from real-world scenarios where hardware and software need to function seamlessly together.
- **Hands-On Workshops:** Organizing more workshops with experts in PCB design tools like Altium Designer and computer vision frameworks like OpenCV can enrich the learning process. Additionally, hosting guest lectures by industry professionals specializing in robotics and automation would further inspire and guide interns.

These improvements will ensure that future interns leave with a deeper skill set, ready to contribute more effectively to the industry.